

Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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SUMMARY

Experienced in Artificial Intelligence, Machine Learning, Large Language Models, and software engineering across research and industry. Skilled in building production-grade software and AI systems, including deep learning models, multi-agent AI pipelines, and full-stack web applications. Research background spans LLM-based software testing, medical AI and applied machine learning.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) **2022 – 2026**
Bachelor of Science in Computer Science and Engineering (CSE) **CGPA: 3.60 / 4.00**
Relevant Coursework: Data Structures and Algorithms, Algorithm Engineering, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Compiler Design, Artificial Intelligence, Machine Learning, Computer Graphics
Research Interests: Large Language Model Systems, AI-Assisted Software Engineering, Applied Machine Learning, Medical AI

PROFESSIONAL EXPERIENCE

- **AI Software Engineer** [[R&D Website](#) | [FABINS](#)] 2026 – Present
Department of Research and Development (R&D) - Saturn Textiles Limited
 - Developing and training computer vision and deep learning models for automated fabric inspection through **FABINS**.
 - Developed full-stack applications with Next.js, React, Spring Boot, and REST APIs, with GitHub Actions for CI/CD.

RESEARCH EXPERIENCE

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing (Undergraduate Thesis)** [[GitHub](#)] 2025 – 2026
Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation
 - Designed a multi-agent LLM framework to automate web test generation and execution from natural-language requirements.
 - Built benchmark datasets and evaluated against ground-truth suites, achieving 84.0% scenario coverage and 90% error detection.
- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [[GitHub](#)] 2026
Python • Deep Learning • Medical AI
 - Built a multi-model chest X-ray reasoning system with a conflict-resolution pipeline to reconcile contradictory model outputs.
 - Designed a framework using confidence calibration and uncertainty-based abstention for reliable predictions.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [[arXiv: 2602.14291](#)] 2026
Python • Speech Processing • Machine Learning
 - Supported data collection and annotation for a large-scale Bangla Automatic Speech Recognition (ASR) and speaker diarization benchmark, enabling evaluation using Word Error Rate (WER) and Diarization Error Rate (DER).

SELECTED PROJECTS

- **MindTrace: AI-Powered Dementia Care Application** [[GitHub](#) | [Feature](#) | [Infrastructure](#)] 2025
Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure
 - Engineered an AI-driven Android app for dementia care with an Android frontend and scalable Spring Boot architecture.
 - Designed accessibility-first UI/UX for cognitively impaired users to reduce cognitive load and improve engagement.
- **Gemma VetCare: AI-Assisted Farming Support System** [[GitHub](#) | [Feature](#)] 2025
Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB
 - Built an AI-assisted Android application delivering real-time veterinary diagnosis and treatment recommendations.
 - Designed RESTful APIs optimized for low-connectivity environments with efficient request handling.
- **TCP Window Scaling Attack** [[GitHub](#)] 2025
Python • TCP/IP • Network Security • Wireshark • Linux
 - Built a virtualized MITM lab to demonstrate ARP poisoning and TCP window scaling attacks between client-server VMs.
 - Implemented attack and defense clients and analyzed TCP traffic using Wireshark and packet captures.
- **Compiler Construction (Flex, Bison, 8086 Codegen)** [[GitHub](#)] 2024
C++17 • Flex • Bison/Yacc • Linux
 - Built a full compiler pipeline for a C-like language, covering lexical analysis, parsing, and semantic analysis.
 - Generated optimized 8086 assembly code with symbol table management and basic optimizations.
- **Smart Gardening Automation** [[GitHub](#) | [Feature](#)] 2024
Arduino • Embedded C++ • Sensors • Bluetooth Communication • Automation
 - Developed a gardening automation system integrating environmental sensing and wireless communication.
 - Implemented automated watering, fertilizer/pesticide cycles, smart shade control, and remote monitoring.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models, Prompt Engineering, Multi-Agent Systems, Computer Vision
Languages: Python, C/C++, Java, JavaScript, Kotlin, SQL
Backend & Full-Stack: Spring Boot, Spring AI, Node.js, Express.js, Next.js, REST APIs, React, Android
Databases: PostgreSQL, MongoDB, Redis
DevOps & Tools: Linux, Git, Docker, Docker Compose, GitHub Actions, Firebase, Azure

ADDITIONAL INFORMATION

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)
Programming Profiles: [NeetCode: mninadmnobo](#) [LeetCode: mninadmnobo](#)
Research Profiles: [ORCID: 0009-0006-2781-6693](#) [ResearchGate: Mohammad Ninad Mahmud Nobo](#)